



Our Mission: Lunar Lyft

Clancy Crawford

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**Cislunar Orbital Live Occupancy
Station and Lander (COLOSAL)
Orbital Hotel: Tranquility
Lunar Base 1: Research Facility
Lunar Base 2: Regolith Mining
Facility**

Assumptions

- Structures on the surface and around the Moon's orbit are already built.
- All three vehicle stages are reusable and refueled between missions.
- All parts of the mission can support safety requirements.
- Unlimited water can be mined on the Moon's poles.
- Resources mined from the Moon is sufficient for our propulsion systems.
- Our payload is comprised of food and supplies for...
 - 20 people (including the operating crew)
 - Cislunar Orbital Station
 - Lunar Habitats
 - Emergency situations

01

Trade Studies

Evaluating Propulsion: Stage 3 - Earth Orbit to Lunar Orbit

- **Model 1:** Hydrogen + Oxygen Liquid Propulsion
- **Model 2:** Thermal Nuclear Propulsion
- **Model 3:** Solar Electric Propulsion
- **Model 4:** Water Electric Propulsion

Alternatives					
	Weight	Model 1	Model 2	Model 3	Model 4
Safety	10	6	1	9	9
Budget Limit	1	7	2	10	9
Vehicle Reusability	7	8	10	7	6
Complexity	5	9	10	5	4
Transportation Time	8	10	9	2	1
System Reliability	6	10	1	9	7
System Efficiency (Power Input Vs Output)	9	7	8	2	7
Failure Likelihood	9	10	3	5	4
Total		461	309	307	310

Ewan Thompson

Evaluating Propulsion: Stage 1 - Ground to Space

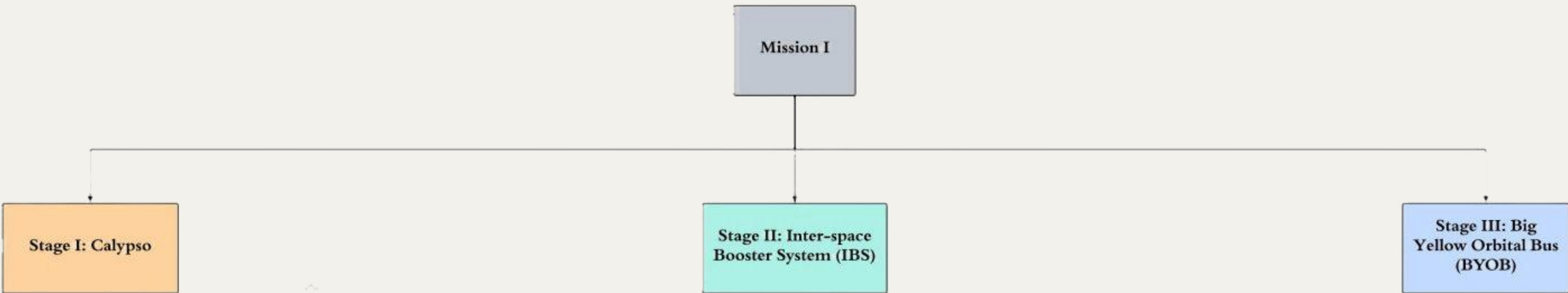
- **Model 1:** Liquid Propulsion
- **Model 2:** Hybrid Propulsion
- **Model 3:** Solid Propulsion
- **Model 4:** Nuclear Propulsion

Evaluation Criterion	Alternative				
	Weight	Model 1	Model 2	Model 3	Model 4
Safety, failure	10	9	9	6	1
Budget Limit	1	7	7	7	2
Vehicle Reusability/Reliability	9	9	7	3	1
Complexity	5	4	3	7	2
Payload Capacity "Mass"	8	5	5	7	10
System Efficiency "Power Input Vs Output"	9	7	8	6	10
Totals		301	287	239	201

02

Work Breakdown Structure

Vehicle Work Breakdown Structure



Work Breakdown Structure Requirements

WBS#	Element	L1	L2	Requirements
1	Mission 1	X		• Mission is completed safely and in a timely manner. • Minimal cost for resupplying stages • Must turn a significant profit
1.1	Stage 1		X	• Must land on launch pad • Must be reusable and refuellable • Must produce enough thrust to carry payload to mesosphere
1.2	Stage 2		X	• Enough fuel to get to LEO orbit, perform TLI, and land in NRHO (docking w COLOSAL) • Must reengage Earths atmosphere and return with minimal damage • Must be reusable and refuellable
1.3	Stage 3		X	• Must reengage Earths atmosphere and return with minimal damage • Life support systems in payload area capable of safely transporting humans • Storage space for tools, food, and amenities

03

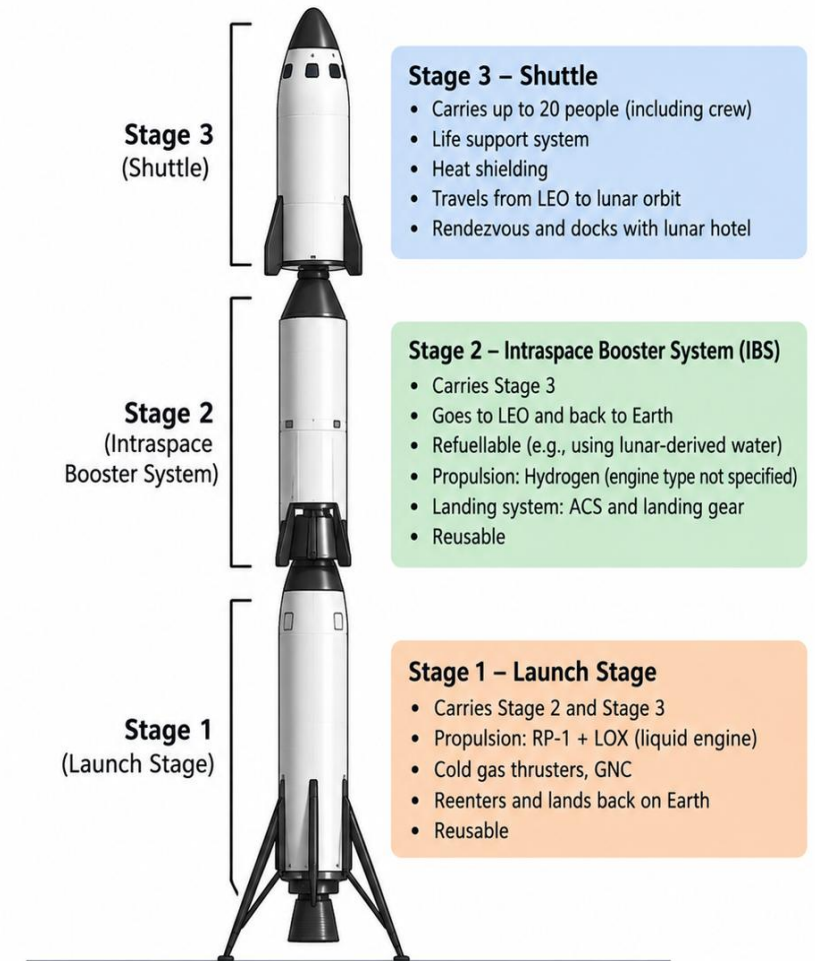
Vehicle Design

Staging

- Stage 1: Ground to Mesosphere
- Stage 2: Low Earth Orbit (LEO)
- Stage 3: LEO to Near Rectilinear Halo Orbit (NRHO)

Mission 1 Rocket

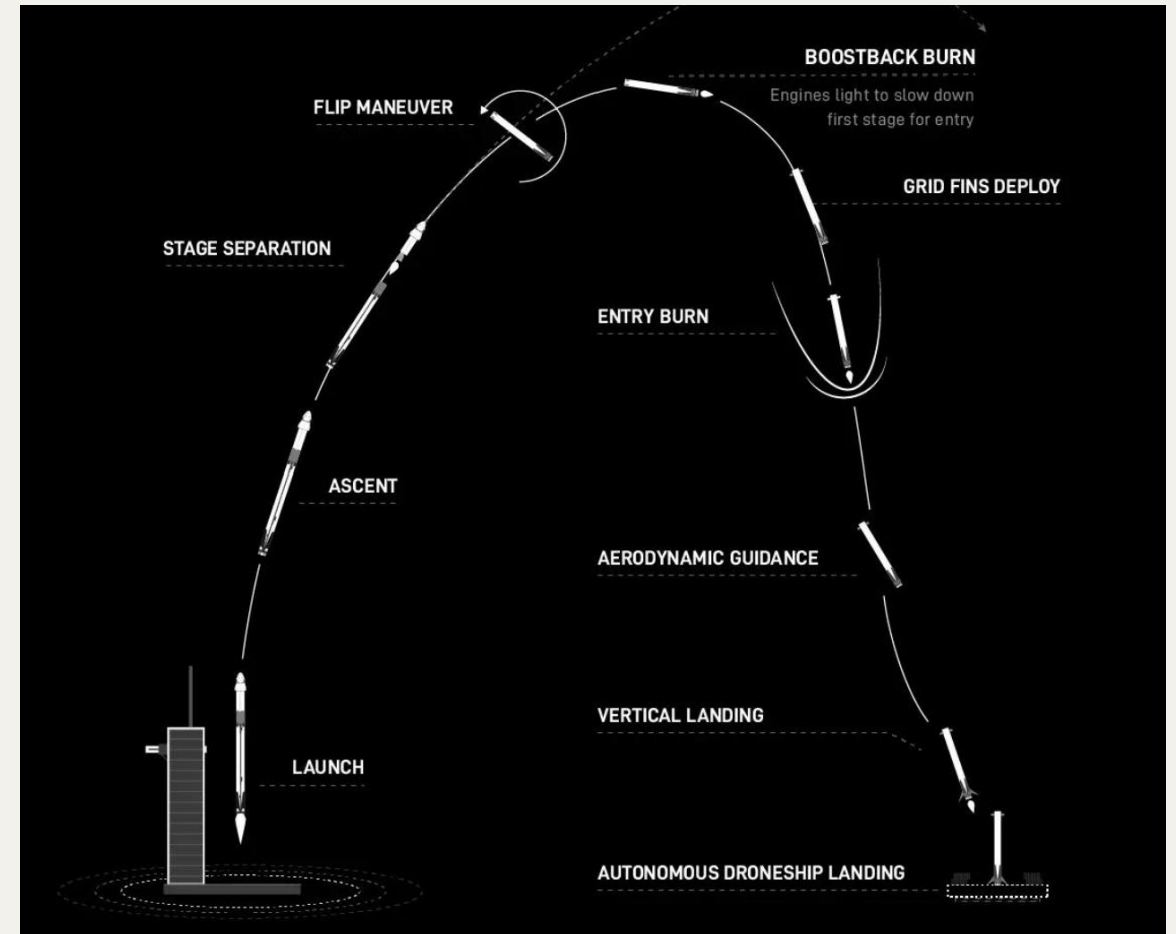
Three Stages. One Destination.



People. Cargo. A Brighter Cislunar Future.

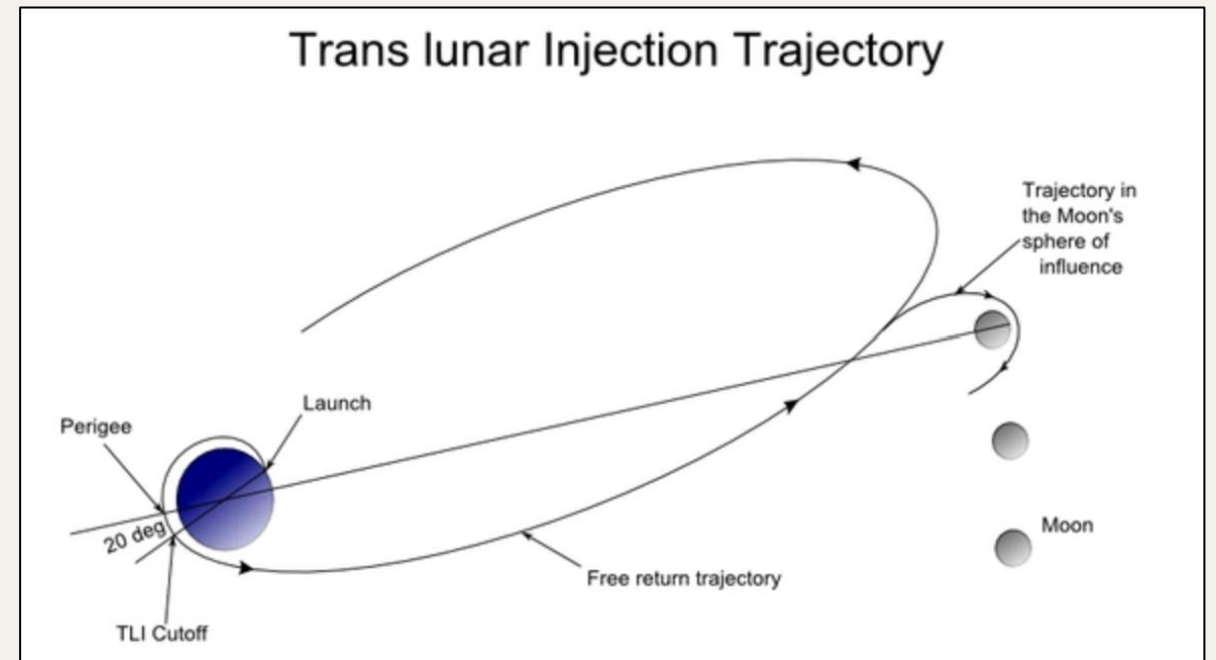
Stage 1: Calypso

- RP-1 and LOX engine provides enough thrust to escape atmosphere
- Stage 1 launches from sea level and takes the remaining stages into the mesosphere
- Stage 1 then enters separation phase and returns to earth at about 80km altitude
- Stage 1 is then recovered, refilled, and reused for future launches



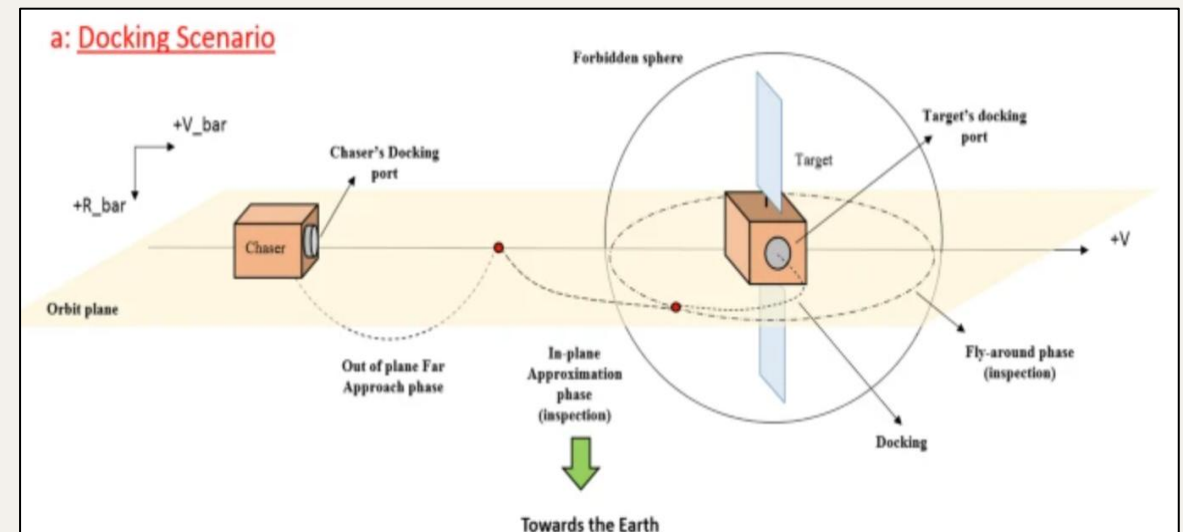
Stage 2: Intra-space Booster System (IBS)

- Maneuvers to park orbit in LEO -> Trans Lunar Injection (TLI)
- After necessary velocity is reached, performs braking maneuver to slow down and fall back to Earth
- Refillable, Hydrogen + LOX fuel



Stage 3: Big Yellow Orbital Bus (B.Y.O.B)

- Shuttle on TLI trajectory after jettisoning spent second stage
- Performs braking maneuver to insert into NRHO and rendezvous with Lunar Orbital Station to dock
- Undocks from station and performs burn to enter a trajectory targeting Earth's upper atmosphere
- Skip Re-entry system
- Food, water, medical supplies, and other resources necessary for operation of lunar bases
- Max occupancy is 20



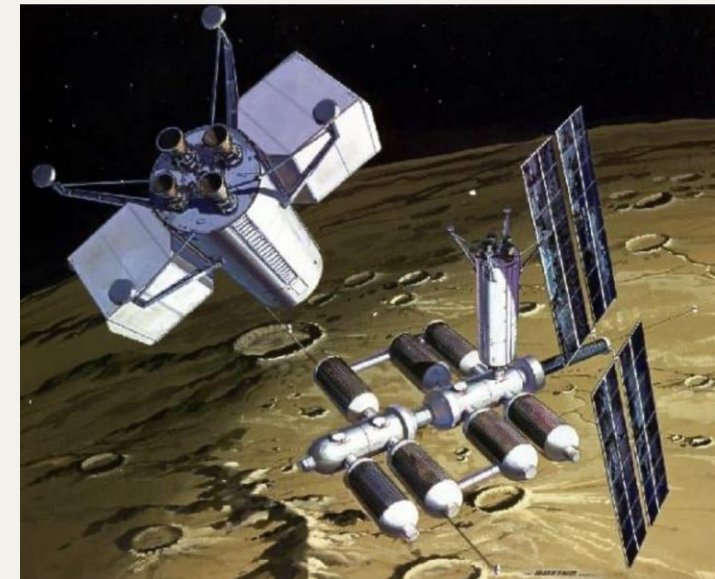
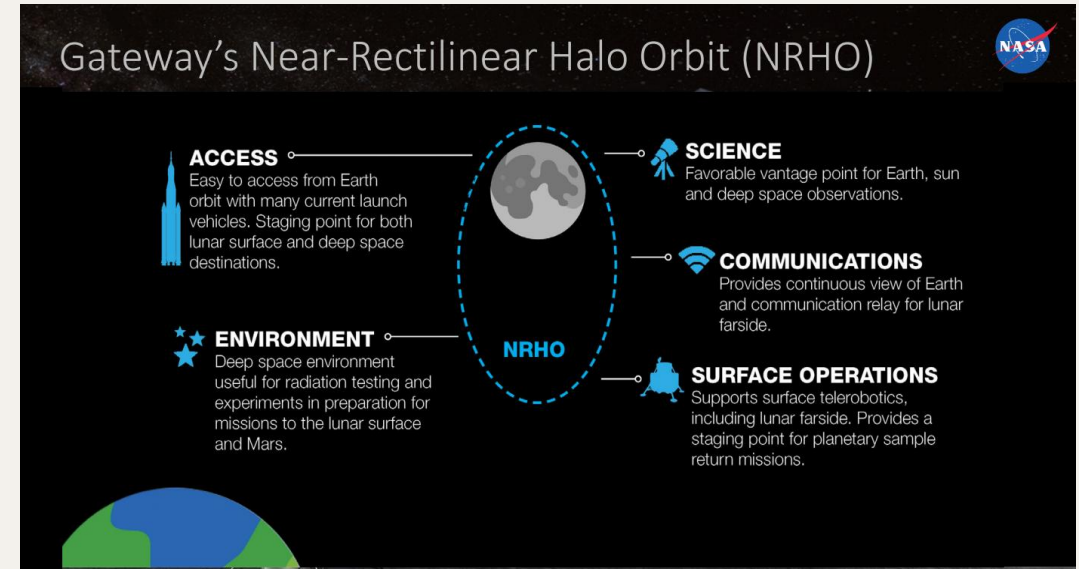
04

Lunar Habitat Design

Cislunar Orbital Live Occupancy Station and Lander (COLOSAL)

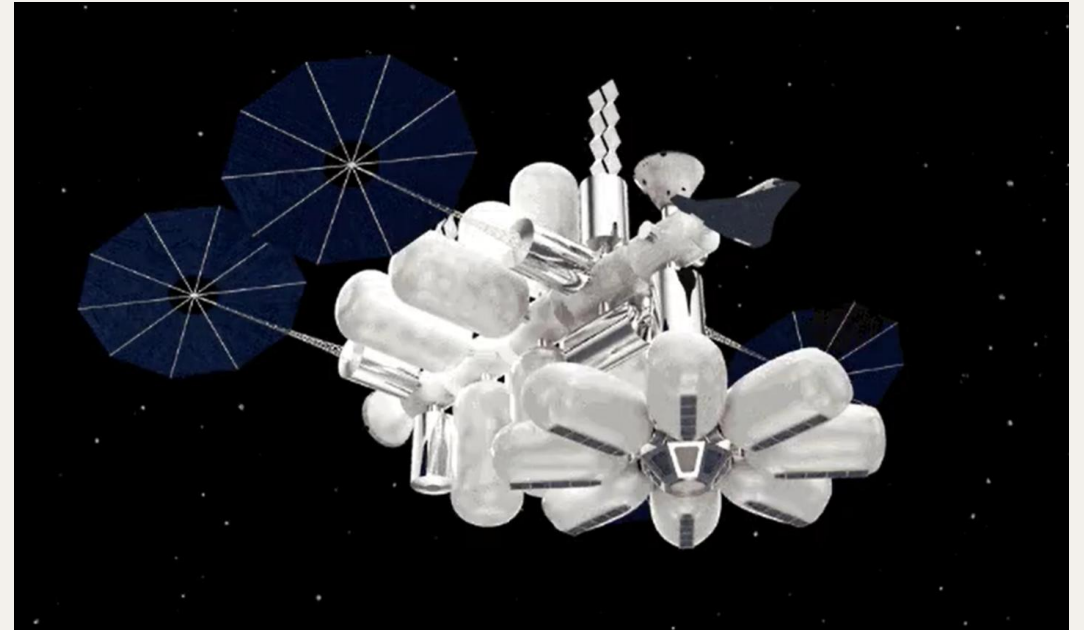
- The CisLunar Orbital Station is a rendezvous point for the final stage of the Launch Vehicle
- NRHO
 - Constant Communication with Earth
 - Fuel Efficiency
 - Periodic access to Moon's surface
- Smaller Lunar Lander descends to Lunar Bases to carry personnel and resources to surface facilities

Lunar Lander Occupancy: 20



Orbital Hotel: Tranquility

- Self-sustaining in NRHO providing amenities and food for 200 guests
- Providing shared living and private luxury suites with constant scenic views of the Earth and the moon
- Maintaining constant communication with the ground station on earth and the other lunar stations



Lunar Base 1: Regolith Mining Facility

- Purpose
 - Provides the lunar base with local sourced materials and resources to support long-term operations.
- Mining and Excavation
 - Collect lunar regolith.
 - Transport material to processing facility.
- Resource Processing
 - Extract water/ice and oxygen.
 - Separate and refine useful materials.
- Base Support
 - Produce construction materials.
 - Provide resources for life support, fuel production, and infrastructure.



Lunar Base 2: Research Facility

- Purpose
 - Researches lunar resources and technologies needed for long-term base operations.
- Research and Development
 - Study lunar regolith and resource extraction.
 - Develop and test lunar-processing technologies.
- Fuel Production
 - Manufacturing machinery for fuel production.
 - Convert extracted water into hydrogen and oxygen
- Technology Testing
 - Test mining, processing and manufacturing equipment.
 - Validate systems before full-scale deployment



Thank you!

Any questions?